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Achieving Data Compatibility over Space and Time: Creating Consistent Geographical Zones

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ABSTRACT

The importance of boundary specification and problems with using arbitrarily defined subnational areas for the collection and dissemination of socioeconomic data are widely recognised. Much research has focused on the modifiable areal unit problem and on custom zone design, but the practical difficulties created by temporal inconsistencies in zonal boundaries have received less consideration. The UK experiences frequent administrative boundary changes causing difficulties in producing comparable statistics through time. Unless a consistent geographical approach is taken with time-series data, it cannot be known whether changes in the relationships between variables collected for areas at different points in time are real or an artefact of boundary changes. After illustrating the nature of the boundary change problems to be overcome, this paper appraises strategies that can be adopted and develops a method by which time-series data can be established on a consistent geographical basis. Copyright © 2003 John Wiley & Sons, Ltd.

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INTRODUCTION

Geographers have long recognised the importance of boundary specification and, since the development of high-performance computing and Geographical Information Systems (GIS), much progress has been made in addressing problems posed by the use of arbitrarily defined subnational areas for the collection and dissemination of socioeconomic data (Blake *et al.*, 2000). Observations that different statistical results are obtained when different geographical boundaries and area subdivisions are used led to a focus on the modifiable areal unit problem (MAUP) (Openshaw and Taylor, 1981) and custom zone design (Openshaw and Rao, 1995). However, practical and generic methods for handling time-series data that have been collected for temporally inconsistent zonal systems have had less consideration.

The UK is subject to more administrative boundary changes over time than the rest of Europe put together (Office for National Statistics, 2000). For example, electoral ward boundaries are regularly adjusted in response to population change to ensure that each local authority has similar councillor/elector ratios (UKSGB, 2000), and the UK local government structure was substantially revised in the late 1990s with the creation of new unitary authorities in some areas (Wilson and Rees, 1999). Even if boundaries do not change, area names and reference codes often vary, with different versions and spellings used across years and between different data suppliers and agencies. These issues are compounded by the large number of different geographies for which data are available, and by technical difficulties because commonly-used administrative and postal geographies do not align. This degree of change and complexity causes problems in the production of comparable statistics over time. Unless a consistent geographical approach is taken with time-series data, it cannot be known whether changes in the relationships between variables collected for areas are real or an artefact of the boundary system in which they were collected.

The research reported in this paper is part of a project to estimate age–sex disaggregated populations for each mid-year 1990–1998 to be used as denominators in standardised mortality ratios (SMRs) at ward level in Government Office Region (GOR) East. However, ward boundaries may have changed incrementally throughout the 1990s, with some altered more substantially with local government restructuring during 1996–98 (Department of the Environment, Transport and the Regions, 1998). In the former case, incremental changes are very difficult to determine. In the latter, the fact that structural changes have occurred can be readily identified as the number of wards and names change, but details about the boundary changes are not easily available. For research through time at electoral ward scale, various boundary-change scenarios may need addressing. These are described below and illustrated in Fig. 1 which shows the 1991 Census wards and 1998 electoral wards for Welwyn–Hatfield, a local authority (LA) district in Hertfordshire.

- In the north of the district, the 1991 wards Welwyn West and Welwyn East have changed to Welwyn South and Welwyn North by 1998. The new wards occupy the same combined area as previously, but the boundary between the two has changed.
- To the south of these in 1991 is Haldens ward, which by 1998 had been subdivided with the creation of an additional ward, Panshanger.
- In the central and southern parts of the district in 1991 are the wards Hatfield East and Brookmans Park and Little Heath. The 1998 boundaries show a substantial part of Hatfield East has been transferred to Brookmans Park and Little Heath.

It is straightforward to identify wards that change names or reference codes, or that wards have been created or abolished. However, when ward names stay the same but their areal extents alter, identification is difficult without detailed maps. Moreover, without a time-series of digital boundaries or regular surveillance of Hansard

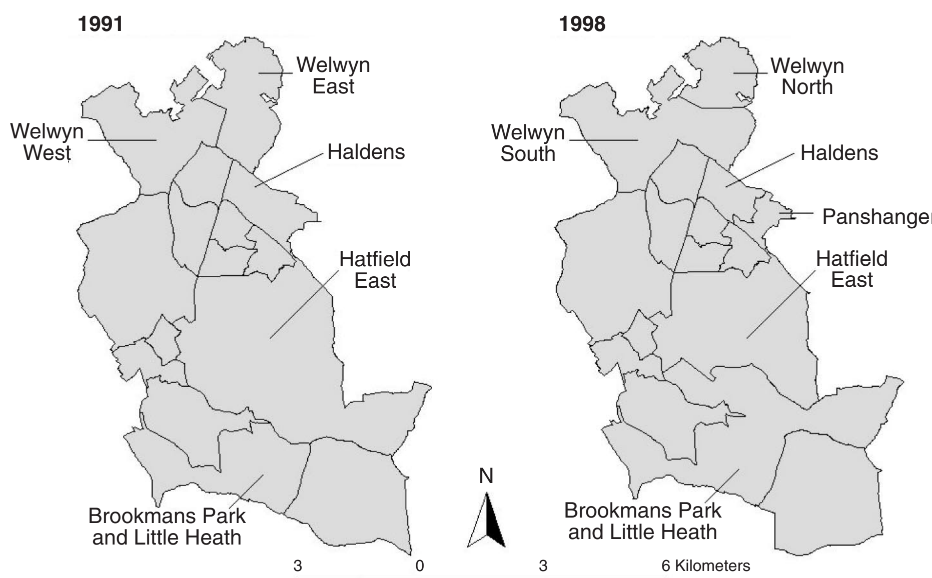

Figure 1. Comparison of ward boundaries, 1991 and 1998, for Welwyn–Hatfield district. *Sources: see Acknowledgements b and g.*

Table 1. Inconsistent geography: male deaths in Welwyn–Hatfield, 1990–98.

Wards	1990	1991	Wards	1992	1993	1994	1995	1996	1997	1998
Brookmans Park and Little Heath	32	25	***Brookmans Park and Little Heath***	23	25	24	28	28	23	18
Haldens	52	56	***Haldens***	38	26	38	38	21	27	31
Handside	37	28	Handside	45	47	29	47	34	35	35
Hatfield Central	27	36	Hatfield Central	38	32	46	27	26	30	33
Hatfield East	25	23	***Hatfield East***	31	24	33	28	27	21	29
Hatfield North	36	31	Hatfield North	33	30	33	32	20	36	34
Hatfield West	21	22	Hatfield West	24	26	31	21	34	28	23
Hollybush	30	37	Hollybush	40	42	30	35	34	31	20
Howlands	25	31	Howlands	27	28	33	30	24	24	35
Northaw	26	18	Northaw	35	24	20	29	34	20	27
–	–	–	***Panshanger***	15	14	12	26	13	15	11
Peartree	33	28	Peartree	45	30	24	31	35	28	32
Sherrards	31	23	Sherrards	24	25	26	22	23	23	25
Welham Green and Redhall	33	29	Welham Green and Redhall	38	41	34	41	43	52	35
Welwyn East	25	14	***Welwyn North***	16	15	9	16	10	13	20
Welwyn West	14	20	***Welwyn South***	27	22	23	19	29	22	15

(1998–2001), it is difficult to know when boundary changes occur. The alterations to wards in Welwyn–Hatfield were made in the month after the 1991 Census (Baker, 1991), thereby creating difficulties interpreting Census outputs within the post-Census geography. To highlight the problem to be overcome, original data from the ward vital statistics (VS) for Welwyn–Hatfield are in presented in Table 1, with ward names highlighted where boundary changes have occurred. Total male deaths are shown for 1990 and 1991, the years preceding the boundary alterations, and for 1992 to 1998, the years following the changes. Evidently, adjustments to cope with the boundary changes are needed so that the VS can be used as annual input data in population estimates or as event counts in SMRs.

The Office for National Statistics (ONS) (1997, 2000) recognised the need for a coordinated approach to geography and consistency in the use of names, codes and references. As a result, mechanisms were put in place in the late 1990s to promote harmonisation within National Statistics including the ONS Geographic Referencing Strategy and the Gridlink project, to create a core set of postcode location data (ONS, 2002a). Various projects within the Economic and Social Research Council (ESRC) Census Development Programme (ESRC, 1999) address many issues relating to boundary changes. For example, Boyle and Feng (2001), Dorling *et al.* (2001) and Martin *et al.* (2002) have adjusted the geographical basis of many census outputs to enable comparisons of census data over time. The Updated UK Area Masterfiles (UUKAM) project allows conversion of data between 1991 Census and various late-1990s postal and administrative geographies (Simpson, 2002; Simpson and Yu, 2003) but not for other years in the 1990s. Since the study period of this research pre-dates the ONS initiatives, is outside the remit of the census projects referred to above, and exceeds the data conversions possible through the UUKAM project, a method must be devised to establish a data time-series on a consistent geographical basis to allow population estimates and SMRs to be compared objectively.

This paper complements Simpson (2002) by giving further examples of data conversions from the literature and by demonstrating how the approach of the UUKAM project can be extended to enable data conversions on an annual basis. Example data, before and after conversion, are presented for an area that has experienced boundary changes over time, and the degree of error involved in the data conversion process is assessed using methods recommended by Simpson (2002). Whilst data preparation needs for calculating ward-level population estimates and SMRs have motivated this aspect of research, the method developed is appropriate to other demographic data, geographical scales and applications. In the research reported here, data-sets will only be used that are nationally consistent, widely available and inexpensive to academics and non-academics such as local government officers and health service professionals. There are many underlying data sources supporting this work; these are referred to in the acknowledgements.

APPROACHES TO CREATING CONSISTENT GEOGRAPHIES

When demographic data are available at two or more points in time and boundary changes have occurred, various approaches can be taken to standardise spatial systems and thereby address the problem of harmonising time-series data on a consistent geographical basis (although, in practice, the approach adopted will largely depend on data availability rather than necessarily being application-driven). One approach to creating a consistent zonal system is to freeze geographical history. This involves fixing the zone system at a point in time and systematically tracking subsequent boundary changes so that data collected for a later system can be adjusted back to the original boundaries. A major disadvantage of freezing geographical history is that the chosen zones become less appropriate to current applications over time. This is particularly important for the distribution of resources, especially for electoral wards, since their boundary changes reflect changing population distributions. For current applications then, an alternative approach is to update data collected for a previous spatial system to a set of contemporary zones. A further solution is to construct designer zones either based on boundaries that are common in different years or through the creation of entirely new geographies. The final strategy involves the use of individual or household-level microdata rather than area-based information, thereby allowing aggregation into the geography most appropriate to the application.

In the sub-sections that follow, various examples are given of approaches that have been taken to adjust data to geographies that are consistent over time. The first set of examples are solutions that have been taken to address specific situations. These generally involve fixing boundaries at a point in time with a variety of techniques used to convert data from other geographies to this system. Next described are more generic strategies that use versatile geographic conversion lookup tables. In the UK, these largely reflect the current state of approaches to creating consistent geographies. The third section outlines a fully flexible system based on geocoded individual-level microdata. These systems exist in some countries, such as Finland, and represent the near future for data conversions in the UK. Until recently, in many countries the tendency has been for *ad hoc* solutions to particular problems. There is a need for more generally applicable solutions.

Application-Specific Solutions to Creating Consistent Geographies

In the European Union, EUROSTAT established the Nomenclature des Unités Territoriales Statistiques (NUTS) geographical system for the production of regional statistics. To monitor trends at the NUTS5 regional level, EUROSTAT adopted a 'freeze history' approach so that data collected for later boundary definitions could be recast back in time (Blake *et al.*, 2000). In contrast, the ONS (2002b) track post-1991 ward boundary amendments and supply lookup tables (LUTs) of changes in Enumeration District (ED)/ward relationships. This 'update to contemporary zones' strategy enables data for these geographies to be converted to later ward definitions (see Rees *et al.*, in press). In addition to the datedness of the freeze history approach, the disadvantages of these approaches are the limited geographies that are available and the continual attention needed to identify boundary changes.

The changing geographies of the decennial UK Censuses have provided the incentive for the reaggregation of data from one census to the geography of another (see Atkins *et al.*, 1993; Dorling and Atkins, 1995; Dorling *et al.*, 2001; Harris *et al.*, 2002) or to recast 1991 Census data to postcensal zones following local government reorganisation (Wilson and Rees, 1998, 1999). In Europe, due to changes in administrative zones (such as municipalities), various studies of internal migration and regional population dynamics have adjusted data to a consistent set of boundaries using LUTs; see, for example, studies of the

Netherlands (Rees *et al.*, 1998) and Spain (Garcia Coll and Stillwell, 2000). Simpson (2002) reviews a variety of methods that have been used to construct LUTs and to calculate the overlap weighting criteria. A difficulty of creating LUTs for geographical data conversion is that direct measures of the weights to be applied to convert between geographies are rarely available.

Areal interpolation (Flowerdew and Green, 1994) can be used to distribute data collected for one set of boundaries across another set of zones, whether the conversion is backwards or forwards in time. In its simplest form the size of the source and target zones is used to weight the source zone values by the area of overlap; a version of this approach was used by White *et al.* (1998) to fit census data from various time periods to 1911 Registration Districts. GIS techniques can be used to construct zones from smaller building-bricks to harmonise the zonal system based on boundaries common in different years. To analyse interregional migration in Australia, Blake *et al.* (2000) assembled small building-bricks to generate designer zones with coincident outer boundaries across a number of censuses. Unfortunately, differences in digitising boundaries undermined the development of automated solutions. Successive boundary overlays had to be manually checked, followed by estimation of the area and size of population involved and a search for the nearest consistent boundary. Apart from being laborious, this approach was a compromise between the competing goals of temporal consistency and the maintenance of the functional integrity of the zones. To analyse population change for 1961–95, Walford (2001) reconstructed the small area geography of Mid-Wales using a combination of GIS techniques, such as point-in-polygon (Martin, 1996) searching of population-weighted centroids with polygon overlay, to create a set of consistent small spatial units. Sophisticated designer zone approaches include the creation of entirely purpose-specific zonal systems so as to represent data more realistically than the existing administrative geographies (Openshaw and Rao, 1995). However, what may be deemed realistic at one time for a phenomenon may be inappropriate at other times, and the resulting geographies can be unrecognisable to the research audience.

Approaches based on areal interpolation, GIS or custom-written software and for more than two time points tend to be complex to apply. To carry out research on relatively large study regions, the necessary digital boundary sets must be available for all time points and geographies relevant to the research. Even when they are, digitising discrepancies through error or changes in legal definitions may indicate boundary changes when none have occurred. Whilst the strategies and techniques in the examples in this section can be re-applied elsewhere, they are effectively application-specific and there is a need for more user-friendly and versatile approaches.

Towards Generic Solutions for Creating Consistent Geographical Zones: Using Postcode Directories

In the UK, LUT directories of unit postcodes linked to administrative area codes produced by the census offices and the Royal Mail have become available through the 1990s. The Postcode-Enumeration District Directory (PCED) for England and Wales and subsequent updates have, for example, enabled users to link data for more recent years back to the census geographies. From 1998, the All Fields Postcode Directory (AFPD) enhanced the PCED information by relating unit postcodes to a range of administrative and statutory zones (Harris *et al.*, 2002). Each record in the AFPD includes a reference to the number of residential addresses in that unit postcode, and a code for each of several geographies within which the postcode mainly lies. This information enables the AFPD to be used to construct 'Geographical Conversion Tables' (GCTs) through which data can be adjusted between zonal systems. The UUKAM project (Simpson, 2002; Simpson and Yu, 2003) used the AFPD to make significant advances in the ease of geographical data conversions by providing web-based, publicly accessible, automated conversions between a wide range of geographies (see http://convert.mimas.ac.uk/). As a result, 'convenient data conversion between geographies is now a reality for non-technicians, with no need for programming and database skills' (Simpson, 2002: 74). Currently, however, a limitation of the AFPD as a GCT is that data can only be converted between 1991 and late 1990s geographies; other years in the 1990s are not available, although the AFPD from 2000 onwards will become available in 2003 for academic use through the ESRC/JISC Census Programme (David Martin, personal communication).

Fully Flexible Solutions to Consistent Geographies: Geocoding Individual or Household-Level Data

The ideal solution to the problems created by boundaries that change over time would be to geocode individual or household-level microdata to discrete addresses, so that data can be reaggregated to reflect each alteration to a boundary system, whether postcode sectors, wards or any other administrative or user-defined zones (Harris *et al.,* 2002). This is the case in various Scandinavian countries where master address lists, continuous population registers and point-based microdata are geocoded to a fine resolution. Aggregations into area data could then be carried out using methods such as the GIS point in polygon technique, accessible to users similarly to the online statistical service in Bradford (Thomasson, 2000).

In the UK, with Ordnance Survey (OS) ADDRESS-POINT (Ordnance Survey, 2002), it is becoming feasible to aggregate data into any arbitrary unit using microdata as the building brick. For the Geographic Referencing Strategy (GRS), all data captured by the ONS will be located as grid-referenced points rather than being assigned to area codes, such as electoral wards, current at the time of data processing (ONS, 2002a, b). Although file sizes will be large and data processing computationally demanding, these points can be aggregated into different boundary sets to provide statistics for whatever area is required (subject to confidentiality thresholds that avoid compromising the data anonymity). These areas may be new geographies specific to an application, but can be the same geographies with boundaries from a different point in time, thereby allowing for the boundary changes occurring in the UK each year.

GEOGRAPHICAL CONVERSION TABLES: A POSTCODE-POINT BUILDING-BRICK METHOD

The need is to standardise the input data required for 1990–98 annual population estimates and SMRs to the 1998 ward geography. An 'update to contemporary zones strategy' is relevant and, given that the ideal point-based microdata are not yet available, can involve GCTs with postcode locations used as building bricks. These GCTs comprise lists of reference codes of the 'source' geography in which data exist with links (via the postcode) to the 'target' zones into which the data are to be converted and an indication of the extent of overlap (expressed as a weight) between the geographies. For further details on the conceptual frameworks involved in GCTs see Simpson (2002) and Wilson and Rees (1998, 1999). As noted above, the UUKAM project (Simpson, 2002) demonstrates data conversion using GCTs and postcode counts to derive apportionment weights between geographies. Each record in the AFPD includes a reference to the number of residential addresses at that postcode and a code for each of several geographies that the postcode mainly lies within. Whilst data conversions between electoral wards for every year from 1990 to 1998 are not available in the UUKAM project, there is potential to adopt a similar approach and extend it through the use of postcode information for additional years.

Postcodes used in GCTs: Approach and Assumptions

The principle then is to disaggregate data for the source wards of each year to a set of unit-postcode building bricks that comprise each ward, and then to reaggregate the data to the 1998 ward target geography using the postcodes that comprise the wards for that year. Figure 2 illustrates the postcode building-brick method being adopted. In a hypothetical source geography, North ward has a constituent set of postcodes, their locations marked × (both black and white). By the target year, East and West wards (in grey) have been created, occupying the same areal extent as the previous North ward. The locations of the postcodes associated with East ward are marked as black ×s and with West ward as white ×s. Population data, say 1000 persons, for North ward can be disaggregated to its 10 constituent postcode building-bricks at 100 persons per postcode (assuming equal weighting for all postcodes). The populations of the target geography can then be constructed from the disaggregated data, with West ward containing six postcode building-bricks and therefore 600 persons, and East ward four building-bricks and 400 persons.

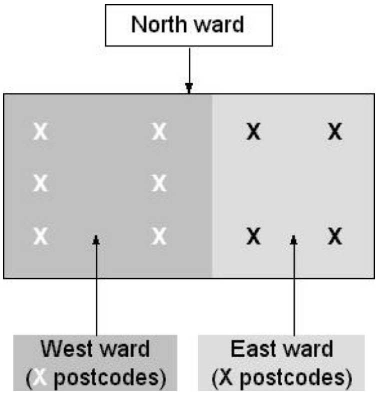

Figure 2. Building-brick data disaggregation–reaggregation approach.

Before reporting the construction of postcode-based GCTs and the method adopted to adjust ward-based data for all years in the study period to be consistent with the 1998 ward geography, it is useful to examine information about postcode locations and the assumptions to be made if postcode-derived apportionment weights are used.

In the UK there are over 1.7 million postcodes covering approximately 25 million addresses. Although created by the Royal Mail for delivering mail, postcodes have evolved since the 1960s as a geographical reference (Raper *et al.*, 1992) with the Chorley Report's (Department of the Environment, 1987) endorsement of the postcode encouraging their widespread use for spatial referencing. Postcodes were recorded for the first time on 1991 Census forms, providing the basis for the creation of the PCED (MIMAS, 1999). Martin (1996) noted that unit postcode locations allow detailed data modelling without the need for digitised boundary information, their main advantage being their small size (typically 14 residential addresses) which offers versatile aggregation into other areal units.

Various postcode-based LUTs available to the academic community enable linkages between geographical areas through time (MIMAS, 1999). In the PCED, for every unit postcode there are details of the ED each postcode is linked with, the number of households in each postcode–ED intersection, and the postcode's National Grid Reference (NGR). The directory was originally compiled with 1991 postcodes and updated for each year from 1995 to 1998. The Central Postcode Directory (CPD) has a single record for each UK postcode, containing the NGR, electoral ward code, date of introduction or termination and postcode user type (small or large). The CPD is available for 1991, 1993 and 1995. The AFPD combines data from various LUTs and is the most recent and comprehensive file linking postcodes with a wide range of geographies. As the PCED gives postcode–ED intersections, a postcode that crosses a boundary can be associated with more than one ward. For the CPD and AFPD, postcodes are allocated on a best-fit basis to the wards existing at the time of file creation. Health professionals have access to similar postcode LUT information in the NHS Postcode Directory (ONS, 2000), and many commercial organisations supply postcode location information (see Martin and Higgs, 1997).

Although unit postcodes each represent a number of addresses, households or large multiple-user buildings, as geographical entities they are taken to be located as points. Postcodes are assigned the NGR of the first address in each postcode (ONS, 2000) as an easting and northing (x–y coordinate) mainly to 100 m resolution (MIMAS, 1999). If postcode point locations are to be used to link source and target geographies and derive intersection weights, various assumptions must be made:

At a point in time, a set of postcodes can be taken to constitute a ward. Figure 1 showed that ward boundaries changed in Welwyn–Hatfield LA district, with Haldens subdivided to create an additional ward, Panshanger. Figure 3 shows the postcodes associated with Haldens ward in 1991 (from the PCED) and the postcodes for Haldens and Panshanger wards in 1998 (from the AFPD). In each year, the constituent postcodes of both wards are largely consistent with the digital ward boundaries, indicating that, at a point in time, a set of postcodes can be taken to constitute a ward.

Residential postcode distribution is a proxy for population distribution. Figure 4 shows the distribution of residential postcode points in GOR East derived from the 1991 PCED. The postcodes are more densely distributed in the more urban areas, suggesting that this assumption is reasonable. There is an urban–rural gradient in the number of households at each postcode, suggesting that if there are more households (and presumably more persons) at some postcodes, there should not be an equal weighting in the disaggregation process. Simpson (2002)

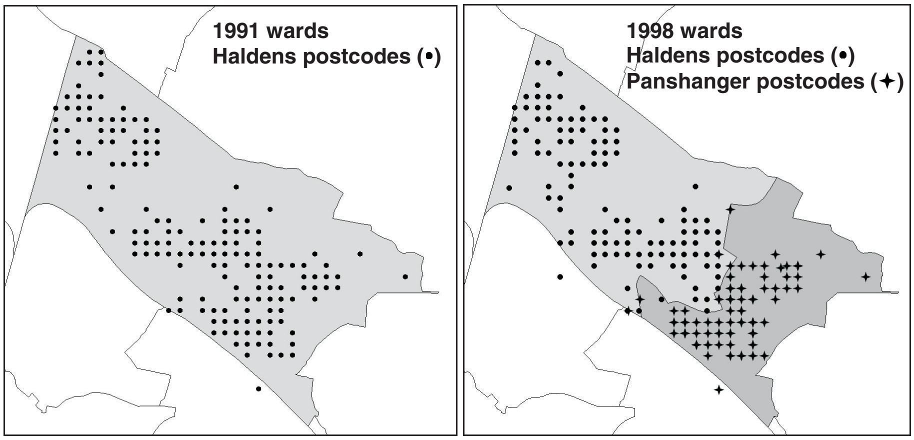

Figure 3. Constituent postcodes of wards in 1991 and 1998. *Sources: see Acknowledgements b, f and g.*

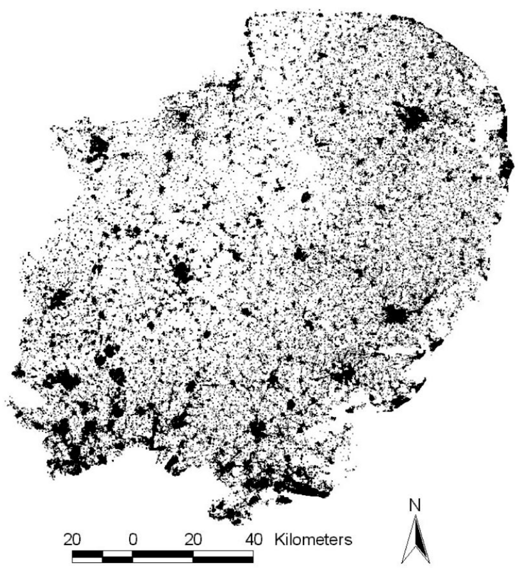

Figure 4. Postcodes as a proxy for population distribution. *Sources: see Acknowledgements f.*

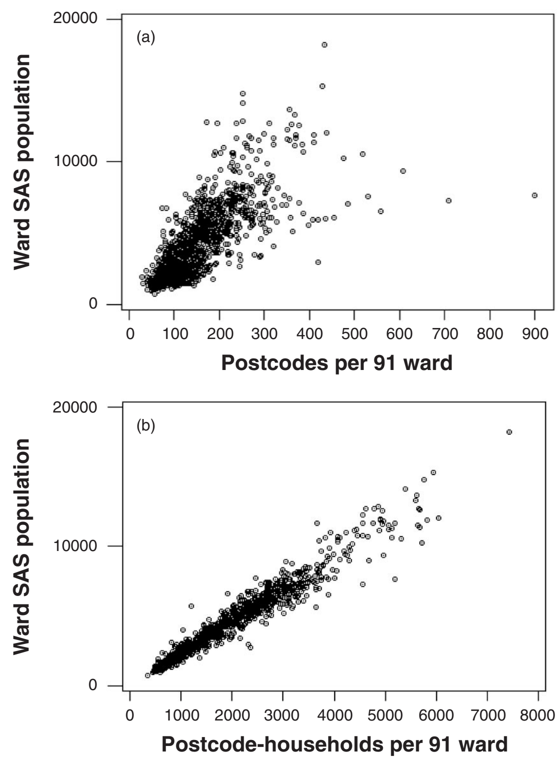

Figure 5. Relationships between postcode-derived counts and ward SAS populations, 1991: (a) numbers of residential postcodes; (b) postcode-household counts. *Sources: see Acknowledgements c and f.*

recommends that the GCT weighting criterion should be correlated with the data being converted, and Simpson and Yu (2003) report that the number of addresses represented by each postcode point is geographically correlated to many social variables (although it should be noted that addresses are a slightly different entity to households). For 1991, Fig. 5a shows a scatterplot of the number of postcodes per ward against the Census Small Area Statistics (SAS) ward populations, a relationship with a correlation of 0.763. Figure 5b illustrates the sum of the postcode-household counts in each ward for 1991 and the SAS populations, a relationship with a stronger correlation of 0.971. Whilst postcode distribution is a proxy for population distribution, data conversions will be enhanced by the use of household counts at each postcode to weight the disaggregations.

The termination and introduction of postcodes is a proxy for population change. The Royal Mail's Postal Address File (PAF) is made available to the ONS each quarter, providing the termination date of redundant postcodes and the date of introduction of new postcodes. Two to three thousand postcodes are terminated each month, and four to five thousand new postcodes are added (Simpson and Yu, 2003). Postcodes are terminated when housing demolitions occur, and new postcodes are allocated when housing estates are constructed and when infill development occurs (Penhale *et al.*, 2000). Although the majority of change is due to 'large user' businesses, the termination and introduction of residential 'small user' postcodes can be taken to be a proxy for residential population change at a sub-ward level.

Constructing the GCTs

In this section, the data preparation necessary to establish annual GCTs will be described. This will be followed by a definition of the calculation of the source zone to target zone intersection weights. To assemble the elements that comprise the GCTs, information is needed for every residential postcode record for each 'year of interest' in the 1990–98 study period. Each record in the GCTs will require:

(i) the source ward code with which each residential postcode is associated;
(ii) a linkage to a target 1998 ward; and
(iii) a count of households at each postcode.

Table 2 shows that the various postcode LUT directories contain similar but inconsistent information, and that LUTs are unavailable for three years of the study period. There are contrasting strengths in the data available in the various postcode directories, since the PCED files contain counts of households at each postcode, whereas the CPDs contain postcode links to wards that are relevant to the year of the file. The AFPD has comprehensive information for 1998. (Only information relevant to this application has been reported here; the original directories contain additional variables: see MIMAS, 1999.) The approach adopted is to use the PCEDs for 1991 and 1995–97 as a basis for the GCTs and enhance them using information from other sources. The steps necessary to construct the GCTs are described below.

(1) *Obtain national postcode LUTs.* The PCED, CPD and AFPD files were downloaded from MIMAS. These will be referred to using a prefix of the LUT name and a two-digit suffix for the year (e.g. CPD95).
(2) *Create GIS point covers for the national LUTs.* For each file, the OS NGRs for each postcode (amended to match the one-metre digital boundary map units) were used to locate

Table 2. Availability of relevant GCT information from postcode directories.

Year	File	Postcode	NGR (xy)	Resolution	Ward code of year	Households/ addresses
1990	✗	✗	✗	✗	✗	✗
1991	CPD	✓	✓	100 m	✓	✗
1991	PCED	✓	✓	100 m	✓	✓
1992	✗	✗	✗	✗	✗	✗
1993	CPD	✓	✓	100 m	✓	✗
1994	✗	✗	✗	✗	✗	✗
1995	CPD	✓	✓	100 m	✓	✗
1995	PCED	✓	✓	100 m	✗	✓
1996	PCED	✓	✓	100 m	✗	✓
1997	PCED	✓	✓	100 m	✗	✓
1998	PCED	✓	✓	100 m	✗	✓
1998	AFPD	✓	✓	100 m	✓	✓

Sources: see Acknowledgements f.

each as a point in the GIS software package ArcView.

(3) *Delete invalid postcodes.* Records were deleted where postcodes were terminated prior to the year of interest and where postcodes were 'large user' business premises, since it is residential postcodes that are required.

(4) *Establish postcode to ward of the year of interest links.* On the PCED91, the CPDs and the AFPD98 each postcode has a link to the ward in existence for the year of the file. Equivalent links are not given on the PCEDs for 1995–97 because the purpose of these directories is to link back to 1991 Census geographies, so there is a need to impute the missing data. This is achieved for the PCED95 using information from the CPD from the same year. For the PCEDs for 1996 and 1997, records are amended from files for adjacent years where the region's ward structure is the same. The PCED96 is enhanced using the CPD95 and the PCED97 using the AFPD98. These enhancements are achieved in two steps.
   (i) Where postcodes are current on each of the files being matched, the ward codes are assumed to be the same. For example, for a postcode that exists in both 1997 and 1998, the ward code from the AFPD98 is assigned to the PCED97.
   (ii) Where postcodes are terminated or introduced, ArcView GIS 'point to nearest point' spatial joins assign the ward attributes of the postcode points of one year to the postcodes in another. This is achieved through radial searches from the coordinates of postcodes in one GIS cover to postcode points in another. For example, a new postcode on the PCED96 file will be assigned the ward code from the nearest postcode in space on the CPD95.

(5) *Create direct links between postcodes of the year of interest and the 1998 ward geography.* For each of the PCEDs (1991 and 1995–97) and CPD93 where postcodes continued to be valid between the file year and 1998, the ward codes from the AFPD were added to the existing PCED or CPD93 postcode records wherever a match could be made. To estimate remaining links, GIS 'point to nearest point' was used whereby postcodes without 1998 ward codes on the PCEDs or CPD93 were assigned the relevant data from the nearest postcode on the AFPD.

(6) *Create household counts for each postcode.* The PCED files for 1991 and 1995–97 already contain the required household counts. For 1993, the only directory available is the CPD. Household counts were therefore estimated for the CPD93 using the PCED95, the postcode directory closest in time with the same ward structure. Record matching was used where postcodes stayed current across the years. For the remaining unmatched postcodes, household counts were assigned to the CPD93 from the nearest postcode in space on the PCED95 using the ArcView 'point to nearest point' technique.

(7) *Allow for years without national LUTs.* Postcode LUTs are not available for three years in the study period. The GCTs from the nearest year with the same ward structure were used.

The results of the above are sets of GCTs for each year 1990–98 that contain ward reference codes to both the source geography for the year in which the data pre-exist, and the 1998 target geography into which the data are to be converted, together with the count of households at each postcode. Additional information retained includes the source and target ward names, with each unit postcode and the easting and northing NGR coordinates. Table 3 has an extract from the GCT for adjusting 1991 data to the 1998 wards, showing postcode linkages that stay the same for Haldens ward and two of those that will enable the transfer of data to the new ward Panshanger.

Calculating GCT Weights for the Data Disaggregation–Reaggregation Process

Data for every ward in each year are to be disaggregated into the ward's constituent postcodes. As a direct link between each source and target ward has been established through each postcode, the data are then reaggregated to the 1998 wards. Weights for the area of overlap between source and target ward geographies are to be derived from counts of postcode-points in combination with the number of households at each postcode. Using the GCTs that have been constructed for each year from 1990 to 1998, the data conversion process is carried out as follows. The source year x ward variable $D^i(x)$ is

Table 3. Geographical conversion table: 1991 to 1998 (extract).

Source ref.	Target ref.	Number of households	Source ward (1991)	Target ward (1998)	Postcode	Easting	Northing
750	861	34	Haldens	Haldens	AL7 2DW	525,900	213,300
750	861	33	Haldens	Haldens	AL7 1NZ	525,000	214,400
750	870	50	Haldens	Panshanger	AL7 2PA	526,500	212,700
750	870	52	Haldens	Panshanger	AL7 2HF	526,200	212,500

disaggregated using the count of households at each unit postcode $H^p(x)$ divided by the total number of households in each source ward i:

$$D^p(x) = D^i(x)\left(\frac{H^p(x)}{\sum_{p\in i} H^p(x)}\right) \quad (1)$$

The estimated postcode variable $D^p(x)$ is reaggregated into the target wards j:

$$D^j(x) = \sum_{p\in j} D^p(x) \quad (2)$$

where:

- D = count of a demographic variable to be converted into a new geography (e.g. births, deaths, electors, etc.);
- x = the index for the year associated with the source geography;
- i = the index for the wards as defined in the source geography;
- j = the index for the wards as defined in the target geography;
- p = the index for the postcodes as defined in the source geography;
- H = the number of households at each postcode;
- $p \in i$ = postcode p is a member of a set of postcodes associated with source ward i; and
- $p \in j$ = postcode p is a member of a set of postcodes associated with target ward j.

Once the source data $D^i(x)$ have been adjusted, the variable $D^j(x)$ is an *estimate* of the original data made for the target geography.

EVALUATING THE POSTCODE-POINT BUILDING-BRICK METHOD FOR GEOGRAPHICAL DATA CONVERSION

In this section the postcode-point building-brick method will be assessed. First, checks are made that boundary changes are being correctly identified using the changes in the constituent postcodes of each ward. Next the output of the method is compared with another data conversion method. Following this, checks are made to determine whether the relationships between postcode-derived counts and populations are maintained when data are converted. Finally, the level of error in the data conversion process is investigated using sample data.

Checking that Boundary Changes are Identified

A test is needed to ensure that the data conversion process correctly identifies the locations and years in which boundary changes occurred. If a ward experienced change between source and target years then the number of postcode-household counts will alter substantially, as it is assumed that, at a point in time, a set of postcodes can be taken to constitute a ward. Figure 3 showed that the set of postcodes associated with Haldens ward reduced with the creation of a new ward, Panshanger. Thus, assuming Haldens had a population of 1000 in 1991, in the conversion process to the 1998 geography this population would be apportioned between Haldens and Panshanger based on the weights derived from the postcode-household counts.

A hypothetical population has been divided equally between the number of source wards in each year, and then converted into the target 1998 ward geography. Once converted (see Fig. 6), counts of less than 1000 represent wards that had reductions in the number of postcode-household counts associated with them by 1998. Counts of more than 1000 will represent wards that have an increased number of postcodes by 1998. Various wards in Welwyn–Hatfield were previously noted as experiencing post-Census boundary

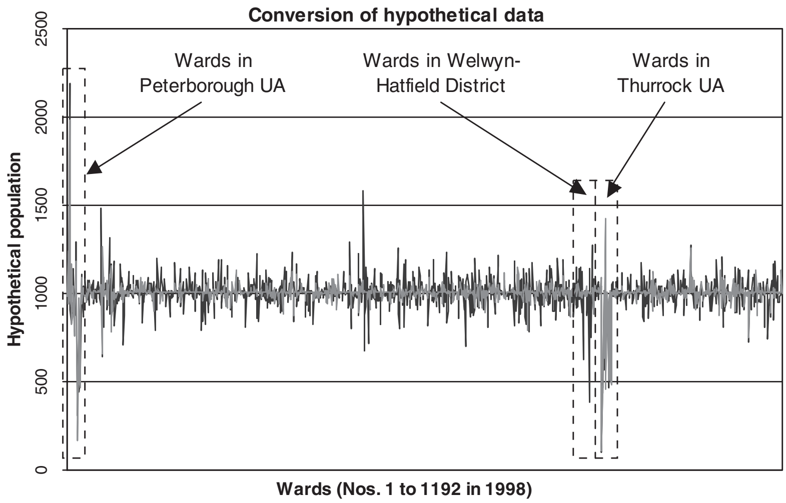

Figure 6. Adjustment of hypothetical populations to 1998 geography.

changes (see Fig. 1). Welwyn South and Brookmans Park and Little Heath, which have larger areal extents than the previous boundary definitions, both have counts above 1000. Similarly, Welwyn North, Haldens and Hatfield East, which all reduced in size, have counts below 1000 because part of the original ward populations have been transferred to wards that have either been created or are larger in the new geography. Other locations with large deviations from 1000 include Peterborough and Thurrock, which both became Unitary Authorities (UAs) by 1998 and experienced substantial changes to their constituent wards. Figure 7 shows the 1991 and 1998 wards in these UAs, demonstrating the extent of the boundary revisions that occurred. Comparisons of mapped postcode locations and 1998 digital ward boundaries (as in Fig. 2) are also consistent with the locations where boundary revisions have occurred. These checks suggest that changes in the constituent postcodes of wards and boundary alterations are being identified in the correct locations.

Comparison with an Alternative Data Conversion Method

The postcode-point program output for 1991 can be compared with the UUKAM project, which offers online conversion between various pairs of geographical zones including conversion from 1991 to 1998 wards. SAS ward populations from the 1991 Census were converted using the project's website (http://convert.mimas.ac.uk/) and plotted against the same population converted using the postcode-point method (see Fig. 8). The scatterplot and correlation of 0.990 demonstrate a high degree of match. This is to be expected since both the postcode-point and UUKAM approaches use LUT information from the AFPD and similar methods to derive the GCT source–target intersection weights. However, variations are inevitable for several reasons:

- The postcode-point building-brick approach uses household counts from the PCED in the year of interest (1991 in this case) to calculate the source–target geography intersection weights, whereas the UUKAM project uses 1998 residential address counts from the AFPD. Disaggregation weights based on data relating to the same year as the input data should be more appropriate since there should be a stronger correlation between postcode-household counts in 1991 and the Census populations than the 1991 populations with 1998 postcode-address counts.
- There is a difference in definition between a household and an address since addresses, in essence, contain households and these can be multiple households. The postcode-point method estimates slightly larger populations in the more urban areas.
- The PCED91 provides the number of resident households in each intersection of postcodes

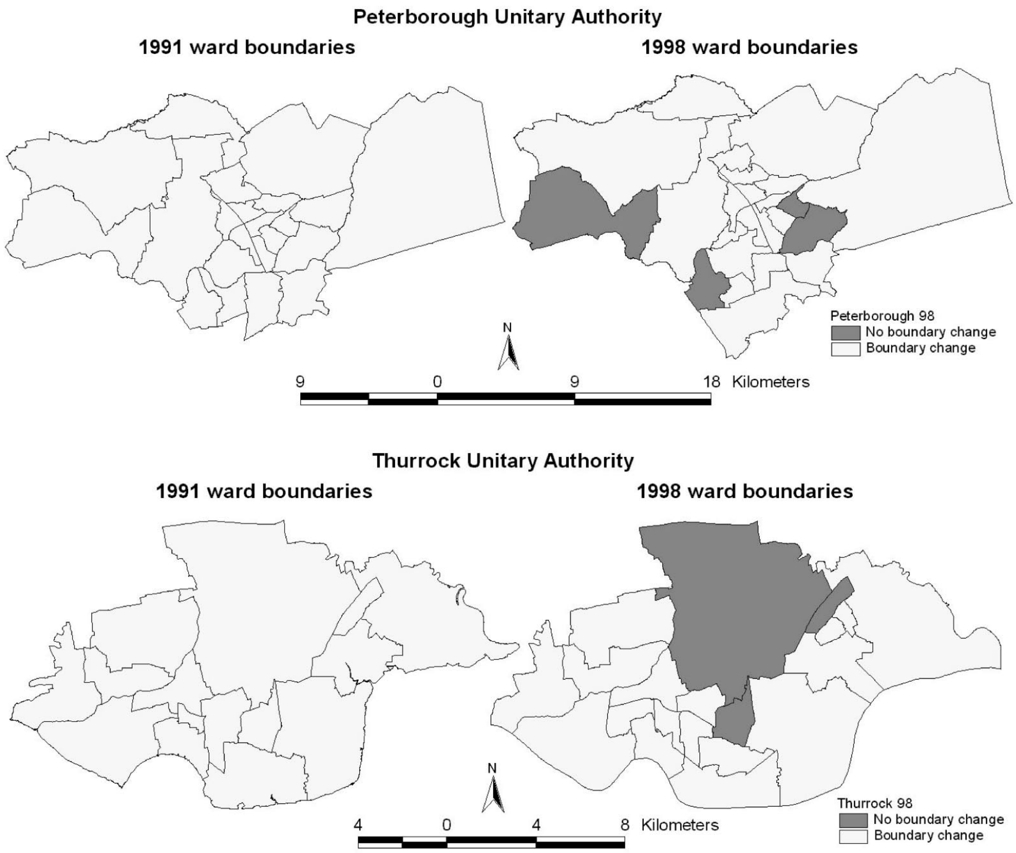

Figure 7. Ward boundaries, 1991 and 1998, Peterborough and Thurrock unitary authorities. *Sources: see Acknowledgements b and g.*

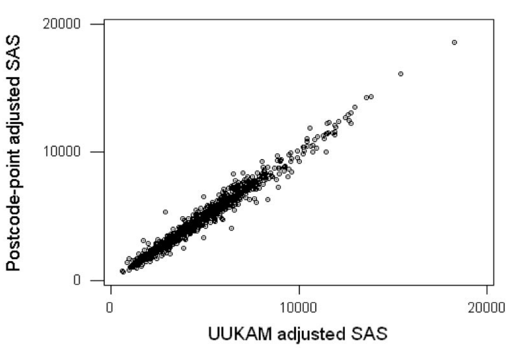

Figure 8. Comparison of postcode-point building-brick and UUKAM conversions.

and EDs so that data can be apportioned to different wards where a postcode crosses a boundary. The AFPD allocates each postcode to a single ward on a best-fit basis. Simpson (2002) notes that error due to the use of whole rather than part postcodes in GCTs exists, but will not be great.

- Links between source and target geographies in the postcode-point approach have been achieved through record matching and information allocated from the nearest postcode. These links may not always be correct and the advantage of the PCED files apportioning postcodes that cross ward boundaries may be lost.
- The region of interest for the postcode-point approach is fixed for GOR East, whereas the UUKAM project uses the national AFPD. Whilst no persons are lost during the data conversion process (i.e. the GCTs in both methods are 'exhaustive': Simpson, 2002: 73), the UUKAM allocates a small number of persons (0.03%) to 1998 wards outside the study region. This implies that about 50 postcodes have different source to target links in the GCTs of the two methods.

Assessing Error in the Data Conversion Process

In the background to the boundary change issues noted in Section 2, various situations were described for wards in Welwyn–Hatfield and illustrated in Fig. 1. Due to the creation of a new ward and changes in the names and areal extents of others just after the 1991 Census, Table 1 showed discontinuities in the time-series of male mortality from the VS for several wards. Table 4 shows the male deaths having been adjusted to the 1998 ward geography using the postcode-point building-brick method described above. It can be seen, for example, that some of the deaths in the original VS for Haldens in 1990 and 1991 have been ascribed to Panshanger, the ward created when Haldens was split. The adjusted deaths data can now be used as both an input to a cohort-component method to estimate denominator populations at risk, and as the mortality event count in SMR calculations for each year from 1990–98.

The data conversions described here will not be error free. The approach is reliant on the quality of the original data in the LUTs obtained from MIMAS, aspects of the unit postcode 100 m NGR resolution, in conjunction with some of steps taken during the construction of the GCTs which may lead to inaccuracies. Simpson (2002) defines two measures to assess the 'goodness of fit' between pairs of geographies involved in the data conversion process: the 'degree of hierarchy' and the 'degree of fit'. The degree of hierarchy (Equation 3) is the proportion of source units that are contained within a single target unit. If the degree of hierarchy is 100%, the source units fully nest within the target units (such as the relationship between electoral wards and LA districts). Simpson and Yu (2003) estimate the degree of hierarchy from the conversion tables for the various postal, census, electoral and other administrative geographies possible using the AFPD. The degree of hierarchy between 1991 and 1998 wards is estimated as 13.8%. This is one of the lowest figures given, but is unsurprising since these source and target geographies are very similar and do not in essence have a hierarchical relationship. For conversion tables extracted from the AFPD, the percentage will represent wards that are estimated to have no boundary change between 1991 and 1998. Table 5 lists the source wards for Welwyn–Hatfield district together with the associated target ward code and maximum conversion weights calculated using the postcode-point building-brick method. Although the majority of maximum weights are over 0.90, only one is equal to 1 and therefore indicates no boundary change. The resulting degree of hierarchy is only 6.67%.

Table 4. Consistent geography: male deaths in Welwyn–Hatfield 1990–98.

Wards	1990	1991	1992	1993	1994	1995	1996	1997	1998
Brookmans Park and Little Heath	35	28	23	24	23	27	27	23	18
Haldens	32	35	38	26	38	38	21	27	31
Handside	37	28	45	47	29	47	34	35	35
Hatfield Central	26	35	37	31	45	27	26	30	33
Hatfield East	25	23	31	24	33	28	27	21	29
Hatfield North	36	31	32	29	32	31	20	36	34
Hatfield West	22	23	25	27	32	22	35	28	23
Hollybush	26	32	39	41	29	34	33	31	20
Howlands	25	31	28	29	33	30	25	24	35
Northaw	26	18	35	24	20	29	34	20	27
Panshanger	20	21	15	14	12	26	13	15	11
Peartree	37	33	45	30	24	31	35	28	32
Sherrards	31	23	24	25	26	22	23	23	25
Welham Green and Redhall	30	27	38	41	34	41	42	52	35
Welwyn North	19	11	16	15	9	16	10	13	20
Welwyn South	21	24	27	22	23	19	29	22	15

Table 5. Degrees of hierarchy and fit for the postcode-point building-brick method for conversion from 1991 to 1998 wards.

1991 wards	Source (i)	Target (j)	Maximum weight (W_{ij})
Brookmans Park and Little Heath	749	860	0.9630
Haldens	750	861	0.6167
Handside	751	862	0.9826
Hatfield Central	752	863	0.9277
Hatfield East	753	864	0.8514
Hatfield North	754	865	0.9571
Hatfield West	755	866	0.8972
Hollybush	756	867	0.8563
Howlands	757	868	0.9599
Northaw	758	869	1.0000
Peartree	759	871	0.9821
Sherrards	760	872	0.9805
Welham Green and Redhall	761	873	0.8697
Welwyn East	762	874	0.7431
Welwyn West	763	875	0.9930

Degree of hierarchy: 6.67% = 100 * (1/15). Degree of fit: 90.53% = 100 * (13.58/15).

The degree of hierarchy is defined as:

$$100 \times \left(\frac{\sum_{ij} (w_{ij}, if(w_{ij} = 1))}{S} \right) \tag{3}$$

where:
i = a source geography unit;
j = a target geography unit;
w_{ij} = a weight, the proportion of the source geography that overlaps the target geography; and
S = the number of source geography units.

The degree of fit sums the largest weight from each source unit and expresses this as a proportion of the number of source units (Equation 4). The degree of fit shows how closely on average the number of addresses (for the AFPD) in a source geography unit matches the number of addresses in its largest overlap with a target unit. This approach recognises that source unit boundaries may be close to the target unit boundaries even when they do not fit them exactly (as in truly hierarchical or one-to-one relationships) and is an important measure of uncertainty. Using the maximum weights in Table 5, the resulting degree of fit for Welwyn–Hatfield is 90.53%. This compares well with the 93% derived overall from the AFPD for 1991 to 1998 ward conversions by Simpson and Yu (2003), considering that five out of fifteen wards in the district experienced boundary changes between 1991 and 1998. It is reasonable to expect that for Peterborough and Thurrock UAs, where extensive boundary changes have occurred, the degree of fit will be lower than that for Welwyn–Hatfield, but the rest of the study region should be higher.

The degree of fit is defined as:

$$100 \times \left(\frac{\sum_{i} (\max w_{st})}{S} \right) \tag{4}$$

where max w_{ij} is the maximum disaggregation weight for a source i to target j overlap.

Simpson (2002) notes that when an independent dataset represents the truth, the approximation involved in data conversion can be measured using the absolute error which can be expressed as a percentage of the true value. Unfortunately a test data-set is not available, but an assessment of the level of uncertainty created by the data conversion process can be made when data are converted between years when boundary changes have not occurred. For GOR East this is the case between 1997 and 1998, so that differences between data before and after conversion will be due to the method. Theoretically, the data-sets should match but there is some variation. The percentage absolute differences between the original and converted data have been calculated; the result can be regarded as error. The median absolute error for the 1192 wards in the region is 0.17, and the median absolute percentage error is 0.51%, both suggesting that the level of error is not great but there is a small amount of variation in almost every ward.

Two phenomena relating to the resolution of the postcode NGRs which can lead to inaccuracies in data conversions should be noted. Firstly, a small number of postcodes are located outside the boundary of the ward with which they are linked in the published directories (as shown in Fig. 3). Secondly, many residential postcodes which lie close together are given the same eastings and northings so the points are stacked. If the extent of an area is being represented by a set of postcode points, these issues will have accuracy implications in both rural and urban areas. The larger extent of rural wards allows more margin for error than in smaller urban wards, where there is greater risk of the postcode being located outside the ward with which it should be associated. Occasionally in rural areas, isolated farms some distance apart may have the same postcode. In these circumstances the NGR may be very inaccurate. It is possible that data might be incorrectly allocated from or to a postcode point that lies the 'wrong' side of a boundary or when stacked postcode locations with the same NGR are associated with different wards. When more than one data record is the nearest point, it cannot be known which record ArcView uses in the LUT attribute table. If incorrect imputation of postcode-ward links are made, error in the data conversion process will result.

There have been various critiques about the use of postcodes as points in space for geographical research (see Gatrell, 1989; Gatrell *et al.*, 1991; Martin, 1992; Heywood, 1997). For the PCED and CPD files, the most consistent criticism relates to aspects of the NGR, particularly the resolution. For example, to improve postcode–ED linkages, Gatrell *et al.* (1991) recommended adding 50 m to both the x and y coordinates, a procedure most advisable if linkages between geographies were established by GIS point-in-polygon operations. Without resorting to expensive purchases, there will not be an improvement in postcode accuracy until the Gridlink-enhanced AFPD (at 1 m resolution) is available to the academic community. Gridlink is a joint venture established by the ONS, the OS, the Royal Mail, the General Register Office for Scotland and the OS of Northern Ireland to develop a co-ordinated approach to the creation of postcode location products. For Gridlink, new methods have been developed for assigning grid references and creating area linkages. Later releases of the AFPD than the version used in this research include postcode NGRs assigned either using the existing 'first address approach' or by the new Gridlink method. The traditional method AFPD will be superseded by the 'Gridlink-enhanced AFPD' in due course. Thus, a problem with using successive postcode directories will be discontinuities in postcode locations from the late 1990s which may be somewhat counterproductive to the creation of consistent geographies over time. The implications of these discontinuities for VS data are noted by the ONS (2002b).

GEOGRAPHICAL DATA CONVERSION: INPUT DATA FOR WARD POPULATION ESTIMATES AND SMRS, 1990–98

Geographical Data Conversion of Population Stocks and Indicators of Change

As noted above, the research reported in this paper is part of a project to estimate ward-level age–sex disaggregated populations for each midyear 1990–98 to be used as denominator populations at risk in SMRs. The principle underlying the estimates is to take a base population derived from the 1991 Census and update this using data contemporary to other years in the study period. (For further details about the population estimation methods, see Rees *et al.*, in press.) The data include the ward-level VS which are used to inform on numbers of births and deaths each year. Electorates are used as an annual ward-level indicator of change in population size. These data have been converted using the method described in this paper from the ward geography of the year of collection to the 1998 geography in which the time-series of results are required. The geographically harmonised deaths from the VS (as in Table 5) are the event counts in the SMRs.

Geographical Data Conversion of Migration Information

In addition to a base population, the 1991 Census also provided 1990–91 inter-ward migration flows which were to be adjusted for the other years in the study period using annual larger area migration data from the National Health Service Central Registry (NHSCR). The 1990–91 in- and out-migration data for each Census ward have been converted to the 1998 ward geography using the postcode-point building-brick method. This approach, however, is less appropriate for interaction flows than for population stock data.

Figure 9a illustrates various inter- and sub-ward migration scenarios for a hypothetical 1991 geography with two wards and six postcode locations shown. A person moves house from postcode a in West ward to postcode *b* in East ward, with another person moving from postcode *d* in East ward to *c* in West ward. These persons are both classified as inter-ward migrants. Since the boundary is not crossed, the person who moves between postcodes *e* and *f* is not an inter-ward migrant. By 1998, a boundary change has occurred with part of West ward transferred to the East (see Fig. 9b). The person who migrated from postcode *a* to *b* is still classified as an inter-ward migrant as they cross the repositioned boundary. However, due to the boundary change, the person moving from postcode d to postcode c is no longer an inter-ward migrant, whereas the person moving between *e* and *f* is now counted as an inter-ward migrant.

Since a change of boundary affects whether or not a person who moves house is counted as a migrant, the out-migration from West ward is 1 in the 1991 geography but 2 in the 1998 geography. The postcode-point building-brick method adjusts the total out-migration from West ward based on the postcode-household counts. Given the reduction in ward size, the out-migration might be underestimated. Similarly, the out-migration from East ward may be overestimated. In locations where substantial boundary changes have occurred, short-distance migration may be poorly adjusted to the new geography. However, it is reasonable to assume that false gains and losses largely offset themselves (as in Fig. 9). The effect on net migration may therefore be small, and inaccuracies at a relatively local level.

Boyle and Feng (2001) addressed the problem of adjusting interaction data for changes in geography by re-estimating data collected for 1981 Census wards to the 1991 ward geography. Their approach used 1981 EDs as a bridge between 1981 and 1991 ward geographies. Flows between and within 1981 EDs were estimated using parameters derived by modelling the ward flows based on origin and destination populations and the distance between them. Point-in-polygon routines were used to identify the 1991 ward within which each 1981 ED centroid fell. The estimated inter-ED flows were then aggregated into the relevant 1991 wards. The method can be applied to other cases where interaction data for one set of zones need to be re-estimated for another overlapping geography, but the computational demands and modelling effort to process interaction data are beyond the scope of this research.

A Time-Series of SMRs For Wards Which Experience Boundary Changes

The geographically harmonised populations at risk and mortality data have been used to calculate SMRs for all wards in GOR East. To alleviate

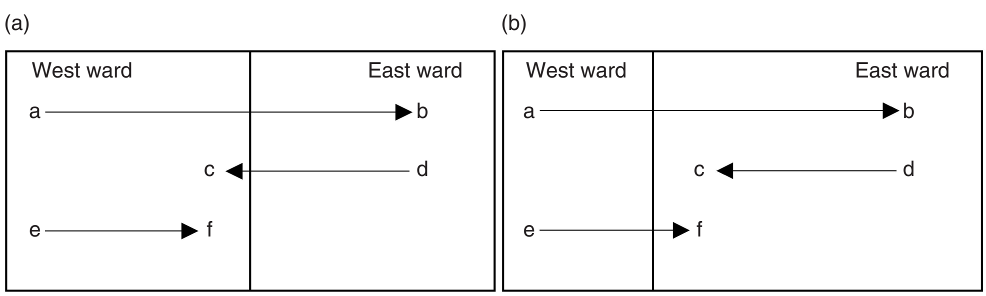

Figure 9. Internal migration and boundary change scenarios: (a) 1991; (b) 1998.

annual fluctuations, the SMRs are three-year rolling averages, apart from the first and last years in the study period which are two-year averages. Figure 10 shows SMRs for 1990–1998 for the wards in Welwyn–Hatfield that experienced boundary changes. Since it was the internal boundaries between various pairs of wards, these are illustrated on the same graphs. In Brookmans Park and Little Heath ward and Hatfield East ward, the SMRs show these wards have mortality experiences that are better than the national situation. Male SMRs show a general improvement over the nine-year period, while female SMRs in both wards stay fairly steady. In Haldens and Panshanger wards, the SMRs are just higher than for England and Wales as a whole at the start of the 1990s. In Haldens, male SMRs show a slight increase in the mid-1990s, but improve by 1998 to below 100. In Panshanger, the male SMRs show a marked improvement up to the mid-1990s, but then rise somewhat. Female SMRs in both Haldens and Panshanger improve at a similar rate in the early 1990s, but then rise back to 1990, levels by 1998. In Welwyn North and Welwyn South, SMRs are advantageous compared with England and Wales. In Welwyn North, male SMRs show a steady improvement through the period, whilst female SMRs remain at a similar level. For Welwyn South, an increase in both male and female SMRs is shown between 1990 and 1998, but despite this, are well below 100.

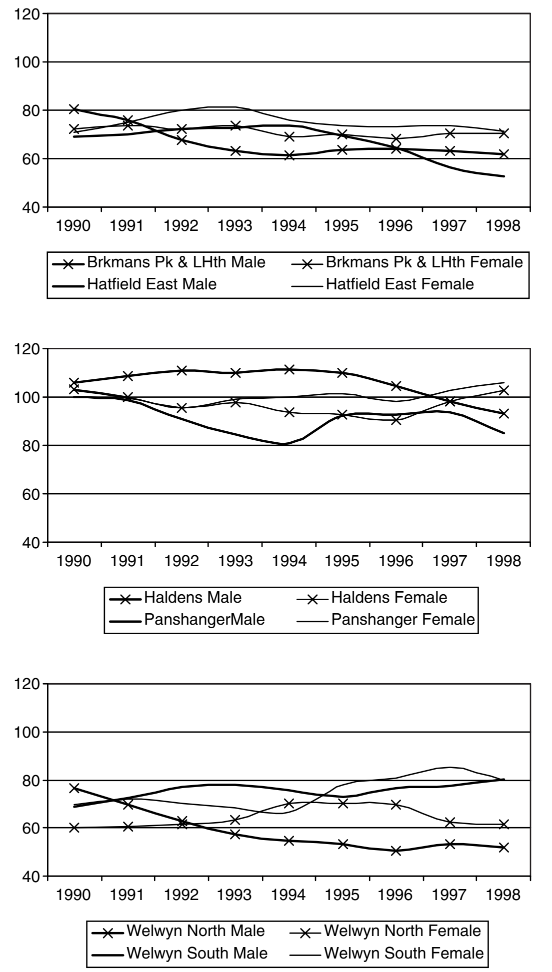

Figure 10. 1990–98 SMR time-series for wards experiencing boundary changes.

CONCLUSION

The harmonisation of geographical zones is essential to a time-series of estimates because area boundaries change over time. From the late 1990s, the availability of the AFPD, the establishment of the ONS Geographic Referencing Strategy and Gridlink, and the dissemination of the 2001 Census in postcode-related outputs (Martin, 2000; ONS, 1999) will enable researchers to establish consistent geographies and link data between different geographies and time periods. However, investigations of time-series data for periods preceding these innovations are hampered by a lack of comprehensive information to tackle the problems posed by boundaries that change over time. Given that various data, such as GIS digital boundaries, either do not exist or are not generally affordable to researchers, this paper has explored how widely available information may be used to establish a consistent ward geography for the period 1990 to 1998. Reviewing the approaches that address the problems posed by the lack of temporal consistency in boundary systems has shown that postcode locations allow detailed data modelling, since their relatively small size allows versatile aggregation into other areal units.

To establish time-series data on a consistent geographical basis, a method has been developed for adjusting population variables. It has been demonstrated that population data for source wards can be disaggregated using geographical conversion tables into the ward's constituent postcodes, and then rebuilt using the constituent postcodes of the target wards for 1998, the latest year for which the input data required for the application are available. The changes in the constituent postcodes of wards are consistent with areas where boundary changes have occurred, and 1991 ward data is estimated for the 1998 ward boundaries similarly to geographical data conversions carried out using the method devised by the UUKAM project. The postcode-point building-brick method uses disaggregation weights derived from the number of households at each postcode from (as near as possible) the same year that the data were collected. The AFPD conversions currently available from 1991 source geographies could be improved by the inclusion of household counts for each postcode from the PCED. Although the differences with postcode-address counts derived from just one year have been shown to be small, the use of a weighting criterion from the source geography year should ensure a stronger correlation with the data being converted. The postcode-point building-brick method has been shown to maintain relationships between ward-level populations and postcode-derived counts, but during the conversion process some error in transferring data between geographical systems is inevitable.

The method developed complements the wide variety of geographical conversions offered by the UUKAM project by demonstrating that conversions for time-series data from other years in the 1990s are feasible. Moreover, by demonstrating the feasibility of creating a consistent geography for a number of years, the method exceeds other techniques reported in the literature which tend just to convert between two zone systems and/or time points. Although the method has been devised in the context of the need to adjust input data for small area population estimates and SMRs, the postcode-point building-brick approach is generically applicable for geographical conversions of population data (assuming a correlation with unit postcode distribution). An electoral ward geography is relevant to this application, but the approach can be used whenever the constituent postcodes of each area of interest and time are known or can be estimated. The approach will be compatible with subsequent releases of postcode directories such as the Gridlink-enhanced AFPD and, whilst the currently available 100 m NGR resolution of the postcodes is less than ideal, the technique will be applicable when more accurately geocoded unit postcode or individual and household-level microdata become widely accessible to the academic community, local government officers and health service professionals.

The recent ONS Geographic Referencing Strategy (ONS, 2002a, b) should ensure that the quality of area-based administrative statistics after 2000 is less affected by changing boundaries. However, as Simpson (2002) notes, the maintenance of data-sets as individual records grid-referenced to points may avoid approximation during aggregation into geographical units, but, for confidentiality as well as practical resource reasons, many data-sets will continue to be held as geographical aggregates. Analysts, particularly outside the ONS and with non-administrative data sources, will still have the challenge of geographical data conversion, and a method such as that developed in this research will not be redundant.

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(b) Digital maps of the Eastern Region are based on data provided by the United Kingdom Boundary Outline and Reference Database for Education and Research Study (UKBORDERS) via the Edinburgh University Data Library (EDINA) with the support of ESRC and the Joint Information Systems Committee of Higher Education Funding Councils (JISC), and boundary material which is Copyright of the Crown, the Post Office and the ED-LINE consortium.
(c) The 1991 Census statistics used in the research are Crown Copyright; the EWCPOP mid-1991 small-area estimates are copyright of the Estimating with Confidence project. Both are made available by the Census Dissemination Unit through the Manchester Information and Associated Services (MIMAS) of Manchester Computing, University of Manchester. The 1991 Census data have been purchased for academic research purposes by ESRC and JISC.
(d) Vital Statistics (VS4) data for wards in the Eastern Region have been extracted from

national files made available by the Office for Population Censuses and Surveys (OPCS) and the Office for National Statistics (ONS). VS4 for 1989–92 have been obtained via MIMAS with permission from the UK Data Archive. VS4 for Eastern Region 1993–94 on paper files purchased direct from the ONS with assistance from Jane Oakley. VS4 for 1995–98 on computer files direct from the Data Archive. The UK Data Archive is jointly funded by the University of Essex (where it is located), ESRC and JISC. Vital Statistics data are Crown Copyright, made available through the ONS and the Data Archive by permission of the Controller of HM Stationery Office.

(e) Electoral data for wards in the Eastern Region have been provided by the ONS with assistance from Gerald Tessier of the Boundary Commission. Numerous local authority and unitary authority Electoral Registrations Units have answered queries about electoral data; their assistance is gratefully acknowledged.

(f) Various lookup tables are made available by the Census Dissemination Unit through the MIMAS service of Manchester Computing, University of Manchester. The Postcode-Enumeration District Directory (PCED) was originally created by the OPCS and has been updated by the ONS. The Central Postcode Directory (CPD) (Postzon) is based on a file originally created by the Department of Transport and has been enhanced by the Post Office, OPCS, GRO(S), Ordnance Survey, the Welsh Office and various local authorities; permission to use the CPD has been given by the UK Data Archive. The All-Fields Postcode Directory (AFPD) is produced by the ONS with information from the ONS, GRO(S), NISRA, the Post Office and the Department of Health. The Updated UK Area Masterfiles ESRC-funded project (H507255164) has re-engineered the AFPD to link census geographies to other administrative geographies. Data in these lookup tables are Crown Copyright, ESRC purchase.

(g) Boundary-Line 1998 digital ward boundaries for the Eastern Region have been made available by special permission of the Ordnance Survey (OS) with assistance from Sallie Payne (then) of the OS and Linda See, School of Geography, University of Leeds. Ordnance Survey is a registered trademark and Boundary-Line is a trademark of the Ordnance Survey, the national mapping agency of Great Britain. OS Boundary-Line data are Crown Copyright. In this research, the 1998 Boundary-Line data will only be used for illustration and checking of results, and not for GIS geometrical techniques, as these boundary sets are unaffordable to most researchers. The availability of archived data for other years from the OS is unclear.

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